

csharp

```
# Rozhranie (interface)
```

```
(1,3): error CS1024: Preprocessor directive expected
```

csharp

**\*\*Rozhranie\*\*** definuje kontrakt obsahujúci určité funkcionality, ktoré musí ne

- rozhranie môže byť deklarované v mennom priestore, v triede alebo v štruktúre
- konvencia je písať ho s prvým veľkým písmenom I
- rozhranie môže obsahovať verejné členy (zvyčajne bez teľa):
  - metódy
  - vlastnosti
  - indexery
  - udalosti
- rozhranie môže byť dediť od jedného rozhrania
- trieda alebo štruktúra môže implementovať jedno alebo aj viacero rozhraní na
- od C# 8.0 môže interface obsahovať vo svojom vnútri aj implementáciu niektorých

```
(1,24): error CS1002: ; expected
```

```
(1,44): error CS1002: ; expected
```

```
(1,66): error CS1044: Cannot use more than one type in a for, using, fixed, or  
declaration statement
```

```
(1,72): error CS1002: ; expected
```

```
(1,90): error CS1002: ; expected
```

```
(1,103): error CS1002: ; expected
```

```
(1,139): error CS1003: Syntax error, '(' expected
```

```
(1,163): error CS1003: Syntax error, ',' expected
```

```
(1,172): error CS1003: Syntax error, ',' expected
```

```
(1,208): error CS1003: Syntax error, ',' expected
```

```
(1,210): error CS1003: Syntax error, ',' expected
```

```
(1,223): error CS1003: Syntax error, ',' expected
```

```
(1,238): error CS1003: Syntax error, ',' expected
```

```
(1,254): error CS1003: Syntax error, ',' expected
```

```
(1,269): error CS1003: Syntax error, ',' expected
```

(1,276): error CS1001: Identifier expected  
(1,276): error CS1001: Identifier expected  
(1,276): error CS1003: Syntax error, ',' expected  
(3,3): error CS1003: Syntax error, ',' expected  
(3,18): error CS1003: Syntax error, ',' expected  
(3,34): error CS1003: Syntax error, ',' expected  
(3,43): error CS1003: Syntax error, ',' expected  
(3,52): error CS1001: Identifier expected  
(3,63): error CS1003: Syntax error, ',' expected  
(3,71): error CS1003: Syntax error, ',' expected  
(3,80): error CS1001: Identifier expected  
(3,80): error CS1003: Syntax error, ',' expected  
(4,3): error CS1003: Syntax error, ',' expected  
(4,16): error CS1003: Syntax error, ',' expected  
(4,25): error CS1003: Syntax error, ',' expected  
(4,33): error CS1003: Syntax error, ',' expected  
(4,49): error CS1003: Syntax error, ',' expected  
(4,50): error CS1001: Identifier expected  
(4,50): error CS1003: Syntax error, ',' expected  
(5,3): error CS1003: Syntax error, ',' expected  
(5,18): error CS1003: Syntax error, ',' expected  
(5,36): error CS1003: Syntax error, ',' expected  
(5,42): error CS1001: Identifier expected  
(5,42): error CS1003: Syntax error, ',' expected  
(5,56): error CS8124: Tuple must contain at least two elements.  
(5,56): error CS1026: ) expected  
(5,62): error CS1018: Keyword 'this' or 'base' expected  
(5,62): error CS1002: ; expected  
(10,13): error CS1002: ; expected  
(10,22): error CS1002: ; expected  
(10,31): error CS1002: ; expected  
(10,48): error CS1002: ; expected  
(11,10): error CS1002: ; expected  
(11,26): error CS1002: ; expected  
(11,45): error CS1002: ; expected  
(11,57): error CS1002: ; expected  
(11,68): error CS1002: ; expected  
(11,82): error CS1002: ; expected  
(12,6): error CS1002: ; expected  
(12,7): error CS1040: Preprocessor directives must appear as the first non-whitespace character on a line

### Príklad vlastného definovaného rozhrania:

```
(1,19): error CS1002: ; expected  
(1,41): error CS1003: Syntax error, ',' expected  
(1,42): error CS1002: ; expected
```

csharp

```
interface ISampleInterface  
{  
    void SampleMethod();  
}  
  
class ImplementationClass : ISampleInterface  
{  
    void ISampleInterface.SampleMethod()  
    {  
        Console.WriteLine("SampleMethod :");  
    }  
}  
  
var obj = new ImplementationClass();  
((ISampleInterface)obj).SampleMethod();
```

SampleMethod :)

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Príklad generického rozhrania [`System.IEquatable<T>`'](<https://docs.microsoft.com/...>)

```
(1,21): error CS1002: ; expected  
(1,32): error CS1056: Unexpected character ''  
(1,53): error CS1525: Invalid expression term ''  
(1,53): error CS1056: Unexpected character ''
```

(1,148): error CS1733: Expected expression

(1,148): error CS1026: ) expected

csharp

```
interface IEquatable<T>
{
    bool Equals(T obj);
}

class Person : IEquatable<Person>, IEquatable<int>
{
    public string FirstName {get; set;}
    public string LastName { get; set; }
    public DateTime Birthdate { get; set; }

    public bool Equals(Person person)
    {
        return (FirstName, LastName, Birthdate) == (person.FirstName, person.La
    }

    public bool Equals(int year)
    {
        return year.Equals(Birthdate.Year);
    }
}

var p1 = new Person { FirstName = "Jan", LastName = "Mrkvicka", Birthdate = Da
var p2 = new Person { FirstName = "Jan", LastName = "Mrkvicka", Birthdate = Da

Console.WriteLine(p1.Equals(p2));
Console.WriteLine(p1.Equals(DateTime.Today.Year));
```

True

True

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```
## Explicitná implementácia rozhrania
```

(1,1): error CS1024: Preprocessor directive expected

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```
Explicitná implementácia používa v názve implementovanej metódy alebo vlastnos
```

(1,26): error CS1002: ; expected  
(1,36): error CS1002: ; expected  
(1,58): error CS1002: ; expected  
(1,71): error CS1002: ; expected  
(1,96): error CS1002: ; expected  
(1,109): error CS1003: Syntax error, ',' expected  
(1,122): error CS1003: Syntax error, ',' expected  
(1,131): error CS1056: Unexpected character '''  
(1,132): error CS1003: Syntax error, ',' expected  
(1,150): error CS1525: Invalid expression term ''  
(1,150): error CS1056: Unexpected character '''  
(1,152): error CS1002: ; expected  
(1,152): error CS7017: Member definition, statement, or end-of-file expected  
(1,165): error CS1002: ; expected  
(1,177): error CS1002: ; expected  
(1,201): error CS1056: Unexpected character '''  
(1,217): error CS1003: Syntax error, ',' expected  
(1,218): error CS1003: Syntax error, ',' expected  
(1,224): error CS1001: Identifier expected  
(1,224): error CS1003: Syntax error, ',' expected  
(1,226): error CS1003: Syntax error, ',' expected  
(1,232): error CS1001: Identifier expected  
(1,232): error CS1003: Syntax error, ',' expected  
(1,246): error CS8124: Tuple must contain at least two elements.  
(1,247): error CS1001: Identifier expected  
(1,247): error CS1056: Unexpected character '''

(1,252): error CS1018: Keyword 'this' or 'base' expected

(1,252): error CS1002: ; expected

csharp

```
class Person : IEquatable<Person>
{
    public string FirstName {get; set;}
    public string LastName { get; set; }
    public DateTime Birthdate { get; set; }

    bool IEquatable<Person>.Equals(Person person)
    {
        return (FirstName, LastName, Birthdate) == (person.FirstName, person.La
    }
}

var p1 = new Person { FirstName = "Jan", LastName = "Mrkvicka", Birthdate = Da
var p2 = new Person { FirstName = "Jan", LastName = "Mrkvicka", Birthdate = Da

Console.WriteLine(p1.Equals(p2)); // Tu sa volá metóda Equals(object) z predka

var p1AsIEquatable = (IEquatable<Person>)p1;
Console.WriteLine(p1AsIEquatable.Equals(p2)); // Tu sa volá implementovaná meto
```

False

True

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Iný príklad:

(1,12): error CS1003: Syntax error, ',' expected  
(1,13): error CS1002: ; expected

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```
class MyData : IEnumerable<int>
{
    private List<int> _numbers = new List<int>();

    public IEnumerator<int> GetEnumerator()
    {
        return _numbers.GetEnumerator();
    }

    IEnumerator IEnumerable.GetEnumerator()
    {
        return GetEnumerator();
    }
}
```

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```
## Rozhrania z .NETu
```

(1,1): error CS1024: Preprocessor directive expected

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```
### Rozhranie `IDisposable`
```

(1,1): error CS1024: Preprocessor directive expected

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Poskytuje priestor na upratanie a uvolnenie vzácných zdrojov, najmä nemanážova

```
``csharp
public interface IDisposable
{
    void Dispose();
}
``
```

```
(1,20): error CS1002: ; expected
(1,33): error CS1002: ; expected
(1,45): error CS1002: ; expected
(1,63): error CS1044: Cannot use more than one type in a for, using, fixed, or
declaration statement
(1,69): error CS1002: ; expected
(1,83): error CS1002: ; expected
(1,83): error CS7017: Member definition, statement, or end-of-file expected
(1,102): error CS1002: ; expected
(1,122): error CS1002: ; expected
(1,136): error CS1002: ; expected
(1,145): error CS1002: ; expected
(1,160): error CS1002: ; expected
(1,167): error CS1002: ; expected
(1,167): error CS7017: Member definition, statement, or end-of-file expected
(1,176): error CS1002: ; expected
(1,183): error CS1002: ; expected
(1,223): error CS1003: Syntax error, ',' expected
(1,243): error CS1001: Identifier expected
(1,245): error CS1031: Type expected
(1,245): error CS1001: Identifier expected
(1,245): error CS1003: Syntax error, ',' expected
(1,249): error CS1002: ; expected
(1,249): error CS7017: Member definition, statement, or end-of-file expected
(3,1): error CS1056: Unexpected character '''
(3,2): error CS1056: Unexpected character '''
(3,3): error CS1056: Unexpected character '''
(3,10): error CS1002: ; expected
(8,1): error CS1056: Unexpected character '''
(8,2): error CS1056: Unexpected character '''
(8,3): error CS1056: Unexpected character '''
```

Ak nejaká trieda implementujete toto rozhranie, mali by sme volať metódu `Dispo:

```
(1,11): error CS1002: ; expected
(1,32): error CS1002: ; expected
(1,48): error CS1044: Cannot use more than one type in a for, using, fixed, or
declaration statement
(1,53): error CS1002: ; expected
(1,60): error CS1002: ; expected
(1,73): error CS1056: Unexpected character '''
(1,74): error CS1002: ; expected
(1,83): error CS1002: ; expected
(1,83): error CS1056: Unexpected character '''
(1,104): error CS1056: Unexpected character '''
(1,105): error CS1003: Syntax error, ',' expected
(1,110): error CS1056: Unexpected character '''
(1,116): error CS1003: Syntax error, ',' expected
(1,136): error CS1003: Syntax error, ',' expected
(1,139): error CS1003: Syntax error, ',' expected
(1,143): error CS1003: Syntax error, ',' expected
(1,148): error CS1003: Syntax error, ',' expected
(1,150): error CS1003: Syntax error, ',' expected
(1,161): error CS1002: ; expected
(1,172): error CS1002: ; expected
(1,181): error CS1002: ; expected
(1,198): error CS1056: Unexpected character '''
(1,199): error CS1002: ; expected
(1,204): error CS1002: ; expected
(1,204): error CS1056: Unexpected character '''
(1,205): error CS7017: Member definition, statement, or end-of-file expected
(1,213): error CS1002: ; expected
(1,213): error CS1056: Unexpected character '''
(1,221): error CS1002: ; expected
(1,221): error CS1056: Unexpected character '''
(1,237): error CS1002: ; expected
(1,261): error CS1003: Syntax error, ',' expected
(1,263): error CS1003: Syntax error, ',' expected
(1,274): error CS1056: Unexpected character '''
(1,275): error CS1026: ) expected
(1,275): error CS1002: ; expected
(1,280): error CS1002: ; expected
```

```
(1,280): error CS1056: Unexpected character ` `
(1,295): error CS1002: ; expected
(1,313): error CS1003: Syntax error, ',' expected
(1,315): error CS1002: ; expected
```

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```
using System.IO;

void WriteTextFile(string path)
{
    // Starší typ zápisu příkazu using - platí v novom vytvorenom bloku
    using (StreamWriter sw = File.CreateText(path))
    {
        sw.WriteLine("Ahoj");
        sw.WriteLine("zo");
        sw.WriteLine("suboru");
    } // Tu sa zavola sw.Dispose() automaticky
}

void ReadTextFile(string path)
{
    // Novší typ příkazu using na riadok bez zložených zátvoriek
    using var sr = File.OpenText(path);
    string s;
    while ((s = sr.ReadLine()) != null)
    {
        Console.WriteLine(s);
    }
} // Tu sa zavola sr.Dispose() automaticky

string fileName = "TextFile.txt";
WriteTextFile(fileName);
ReadTextFile(fileName);
```

Ahoj

zo

suboru

csharp

Príklad pre uvoľnenie zdrojov s použitím `Dispose()` metódy a odporúčaného vzor

```
(1,13): error CS1002: ; expected
(1,31): error CS1002: ; expected
(1,42): error CS1056: Unexpected character ''
(1,43): error CS1002: ; expected
(1,52): error CS1002: ; expected
(1,52): error CS1056: Unexpected character ''
(1,63): error CS1002: ; expected
(1,99): error CS1002: ; expected
(1,99): error CS7017: Member definition, statement, or end-of-file expected
(1,115): error CS1002: ; expected
(1,138): error CS1002: ; expected
(1,150): error CS1002: ; expected
(1,167): error CS1519: Invalid token '' in a member declaration
(1,167): error CS1056: Unexpected character ''
(1,176): error CS1525: Invalid expression term 'bool'
(1,181): error CS1002: ; expected
(1,181): error CS1056: Unexpected character ''
(1,182): error CS7017: Member definition, statement, or end-of-file expected
```

csharp

```
using System;
using System.ComponentModel;

public class MyResource : IDisposable
{
    // Smerník na externý nemanážovaný zdroj, ktorý po skončení používania tre
    private IntPtr handle;
    // Príklad na iný manažovaný zdroj, ktorý je vhodné uvoľniť skôr
```

```

private Component component = new Component();
private bool disposed = false;

public MyResource(IntPtr handle)
{
    this.handle = handle;
}

public void Dispose()
{
    Dispose(true);
    // Zabraňuje, aby sa duplicitne nevolal Garbage Collector
    GC.SuppressFinalize(this);
}

protected virtual void Dispose(bool disposing)
{
    if (!this.disposed)
    {
        if (disposing)
        {
            component.Dispose();
        }

        CloseHandle(handle);
        handle = IntPtr.Zero;

        disposed = true;
    }
}

[System.Runtime.InteropServices.DllImport("Kernel32")]
private extern static Boolean CloseHandle(IntPtr handle);

// C# finalizér (deštruktor) – ak sa nezavolala metóda Dispose, Garbage Co
// Je to ale v nedefinovanom čase a môže to byť už neskoro
~MyResource()
{
    Dispose(false);
}
}

```

csharp

```
### Rozhrania `IComparable` a `IComparable<T>`
```

(1,1): error CS1024: Preprocessor directive expected

csharp

```
public class Temperature : IComparable<Temperature>
{
    public int CompareTo(Temperature obj)
    {
        if (obj == null)
            return 1;

        return this.Fahrenheit.CompareTo(obj.Fahrenheit);
    }

    public double Fahrenheit { get; set; }

    public double Celsius
    {
        get => (Fahrenheit - 32) * (5.0/9);
        set => Fahrenheit = (value * 9.0/5) + 32;
    }
}
```

csharp

```
### Rozhrania `IEnumerable` a `IEnumerable<T>`
```

(1,1): error CS1024: Preprocessor directive expected

csharp

Umožňujú prechádzať a získavať hodnoty v kolekcii sekvenčne. Vďaka nim je možné  
Definícia rozhraní `IEnumerable` a `IEnumerable<T>` v knižnici .NETu:

```
```csharp
public interface IEnumerable
{
    IEnumerator GetEnumerator();
}

public interface IEnumerable<out T> : IEnumerable
{
    IEnumerator<T> GetEnumerator();
}
```
```

Ak si rozhranie `IEnumerable` alebo `IEnumerable<T>` implementujete, umožníte

```
```csharp
public interface IEnumerator
{
    bool MoveNext();
    object Current { get; }
    void Reset();
}

public interface IEnumerator<out T> : IDisposable, IEnumerator
{
    T Current { get; }
}
```
```

Následne volá metódu `MoveNext()`, ktorou sa posunie vpred na prvý záznam a al

```
(1,21): error CS1002: ; expected
(1,32): error CS1002: ; expected
(1,42): error CS1002: ; expected
(1,68): error CS1003: Syntax error, '(' expected
(1,75): error CS1003: Syntax error, ',' expected
```

(1,88): error CS1003: Syntax error, ',' expected  
(1,98): error CS1003: Syntax error, ',' expected  
(1,106): error CS1003: Syntax error, ',' expected  
(1,122): error CS1003: Syntax error, ',' expected  
(1,137): error CS1003: Syntax error, ',' expected  
(1,138): error CS1003: Syntax error, ',' expected  
(2,20): error CS1056: Unexpected character ``'  
(2,21): error CS1003: Syntax error, ',' expected  
(2,32): error CS1001: Identifier expected  
(2,32): error CS1056: Unexpected character ``'  
(2,34): error CS1003: Syntax error, ',' expected  
(2,36): error CS1001: Identifier expected  
(2,36): error CS1056: Unexpected character ``'  
(2,37): error CS1003: Syntax error, ',' expected  
(2,51): error CS1001: Identifier expected  
(2,51): error CS1056: Unexpected character ``'  
(2,53): error CS1003: Syntax error, ',' expected  
(2,64): error CS1003: Syntax error, ',' expected  
(2,65): error CS1003: Syntax error, ',' expected  
(2,69): error CS1001: Identifier expected  
(2,69): error CS1003: Syntax error, ',' expected  
(4,1): error CS1056: Unexpected character ``'  
(4,2): error CS1056: Unexpected character ``'  
(4,3): error CS1056: Unexpected character ``'  
(4,4): error CS1003: Syntax error, ',' expected  
(4,10): error CS1001: Identifier expected  
(4,10): error CS1026: ) expected  
(4,10): error CS1002: ; expected  
(14,1): error CS1056: Unexpected character ``'  
(14,2): error CS1056: Unexpected character ``'  
(14,3): error CS1056: Unexpected character ``'  
(16,7): error CS1002: ; expected  
(16,17): error CS1002: ; expected  
(16,17): error CS1056: Unexpected character ``'  
(16,29): error CS1002: ; expected  
(16,29): error CS1056: Unexpected character ``'  
(16,37): error CS1002: ; expected  
(16,37): error CS1056: Unexpected character ``'  
(16,52): error CS1525: Invalid expression term ''  
(16,52): error CS1002: ; expected  
(16,52): error CS1056: Unexpected character ``'  
(16,68): error CS1002: ; expected  
(16,68): error CS7017: Member definition, statement, or end-of-file expected  
(16,90): error CS1002: ; expected  
(16,102): error CS1002: ; expected

(16,123): error CS1044: Cannot use more than one type in a for, using, fixed, or declaration statement

(16,127): error CS1002: ; expected

(16,138): error CS1002: ; expected

(16,144): error CS1002: ; expected

(16,151): error CS1003: Syntax error, '(' expected

(16,151): error CS1525: Invalid expression term '.'

(16,157): error CS1515: 'in' expected

(16,157): error CS0230: Type and identifier are both required in a foreach statement

(16,169): error CS1026: ) expected

(16,178): error CS1002: ; expected

(16,195): error CS1056: Unexpected character ''

(16,196): error CS1002: ; expected

(16,211): error CS1002: ; expected

(16,211): error CS1056: Unexpected character ''

(16,212): error CS7017: Member definition, statement, or end-of-file expected

(16,227): error CS1002: ; expected

(16,249): error CS1002: ; expected

(16,276): error CS1056: Unexpected character ''

(16,277): error CS1002: ; expected

(16,288): error CS1002: ; expected

(16,288): error CS1056: Unexpected character ''

(16,290): error CS1525: Invalid expression term '/'

(16,292): error CS1525: Invalid expression term ''

(16,292): error CS1002: ; expected

(16,292): error CS1056: Unexpected character ''

(16,307): error CS1525: Invalid expression term ''

(16,307): error CS1002: ; expected

(16,307): error CS1056: Unexpected character ''

(16,308): error CS7017: Member definition, statement, or end-of-file expected

(18,1): error CS1056: Unexpected character ''

(18,2): error CS1056: Unexpected character ''

(18,3): error CS1056: Unexpected character ''

(18,10): error CS1002: ; expected

(30,1): error CS1056: Unexpected character ''

(30,2): error CS1056: Unexpected character ''

(30,3): error CS1056: Unexpected character ''

(32,16): error CS1002: ; expected

(32,23): error CS1002: ; expected

(32,23): error CS1056: Unexpected character ''

(32,34): error CS1002: ; expected

(32,34): error CS1056: Unexpected character ''

(32,35): error CS7017: Member definition, statement, or end-of-file expected

(32,47): error CS1002: ; expected

(32,61): error CS1002: ; expected  
(32,69): error CS1002: ; expected  
(32,78): error CS1002: ; expected  
(32,91): error CS1044: Cannot use more than one type in a for, using, fixed, or declaration statement  
(32,97): error CS1002: ; expected  
(32,105): error CS1002: ; expected  
(32,105): error CS1056: Unexpected character ``'  
(32,110): error CS1002: ; expected  
(32,110): error CS1056: Unexpected character ``'  
(32,111): error CS7017: Member definition, statement, or end-of-file expected  
(32,123): error CS1002: ; expected  
(32,141): error CS1002: ; expected  
(32,151): error CS1002: ; expected  
(32,161): error CS1002: ; expected  
(32,172): error CS1044: Cannot use more than one type in a for, using, fixed, or declaration statement  
(32,179): error CS1002: ; expected  
(32,193): error CS1056: Unexpected character ``'  
(32,194): error CS1002: ; expected  
(32,199): error CS1002: ; expected  
(32,199): error CS1056: Unexpected character ``'  
(32,200): error CS7017: Member definition, statement, or end-of-file expected  
(32,213): error CS1044: Cannot use more than one type in a for, using, fixed, or declaration statement  
(32,216): error CS1002: ; expected  
(32,223): error CS1002: ; expected  
(32,223): error CS1056: Unexpected character ``'  
(32,228): error CS1002: ; expected  
(32,228): error CS1056: Unexpected character ``'  
(32,229): error CS7017: Member definition, statement, or end-of-file expected  
(32,243): error CS1002: ; expected  
(32,259): error CS1002: ; expected  
(32,270): error CS1002: ; expected  
(32,270): error CS1056: Unexpected character ``'  
(32,278): error CS1002: ; expected  
(32,278): error CS1056: Unexpected character ``'  
(32,279): error CS7017: Member definition, statement, or end-of-file expected  
(32,288): error CS1002: ; expected  
(32,296): error CS1003: Syntax error, '(' expected  
(32,310): error CS1515: 'in' expected  
(32,314): error CS1026: ) expected  
(32,330): error CS1002: ; expected  
(32,348): error CS1044: Cannot use more than one type in a for, using, fixed, or declaration statement

```
(32,352): error CS1002: ; expected
(32,359): error CS1002: ; expected
(32,359): error CS1056: Unexpected character ''
(32,370): error CS1002: ; expected
(32,370): error CS1056: Unexpected character ''
(32,388): error CS1056: Unexpected character ''
(32,389): error CS1002: ; expected
(32,394): error CS1002: ; expected
(32,394): error CS1056: Unexpected character ''
```

csharp

```
// Vygeneruje 6 hodnôt od nuly { 0, 1, 2, 3, 4, 5 };
//IEnumerable<int> numbers = Enumerable.Range(0, 6);

// Alebo môžeme použiť:
//int[] numbers = { 0, 1, 2, 3, 4, 5 };
List<int> numbers = new List<int>() { 0, 1, 2, 3, 4, 5 };

Console.WriteLine("Výpis čísiel automaticky cez foreach:");
foreach (var number in numbers)
{
    Console.WriteLine(number);
}

Console.WriteLine("Výpis čísiel manuálnym volaním enumerátora:");
IEnumerator<int> enumerator = numbers.GetEnumerator();
while (enumerator.MoveNext())
{
    Console.WriteLine(enumerator.Current);
}
```

Výpis čísiel automaticky cez foreach:

1

2

3

4

5

VÄ"pis ÄŦÄsiel manuÄ`lnym volanÄm enumerÄ`tora:

0

1

2

3

4

5

csharp

#### Klúčové slovo yield

(1,1): error CS1024: Preprocessor directive expected

csharp

`yield` - indikuje, že metóda, operátor alebo get časť, v ktorých sa môže nachádzať návratový typ musí byť: `IEnumerable`, `IEnumerable<T>`, `IEnumerator`,  
- ich deklarácii nesmie byť použitý parameter s kľúčovým slovom `in`, `ref`  
- nedá sa použiť v anonymných metódach a metódach označených ako `unsafe`

Zavedené od C# 2, odpadá nutnosť manuálneho vytvárania enumerátora (vlastnej triedy)

Použitie:

- `yield return` výraz; - vráti jeden prvok v jednom čase
- `yield break`; - ukončí iterovanie

(1,1): error CS1056: Unexpected character `''`

(1,7): error CS1002: ; expected

(1,7): error CS1056: Unexpected character `''`

(1,9): error CS1056: Unexpected character 'â€‘'

(1,19): error CS1002: ; expected

(1,19): error CS7017: Member definition, statement, or end-of-file expected

(1,32): error CS1044: Cannot use more than one type in a for, using, fixed, or declaration statement

(1,41): error CS1002: ; expected

(1,51): error CS1002: ; expected

(1,55): error CS1002: ; expected  
(1,55): error CS7017: Member definition, statement, or end-of-file expected  
(1,67): error CS1002: ; expected  
(1,75): error CS1002: ; expected  
(1,84): error CS1002: ; expected  
(1,84): error CS7017: Member definition, statement, or end-of-file expected  
(1,99): error CS1002: ; expected  
(2,7): error CS1002: ; expected  
(2,21): error CS1002: ; expected  
(2,29): error CS1003: Syntax error, ',' expected  
(2,31): error CS1056: Unexpected character ``'  
(2,32): error CS1002: ; expected  
(2,43): error CS1002: ; expected  
(2,43): error CS1056: Unexpected character ``'  
(2,44): error CS7017: Member definition, statement, or end-of-file expected  
(2,46): error CS1056: Unexpected character ``'  
(2,61): error CS1525: Invalid expression term ''  
(2,61): error CS1002: ; expected  
(2,61): error CS1056: Unexpected character ``'  
(2,62): error CS7017: Member definition, statement, or end-of-file expected  
(2,64): error CS1056: Unexpected character ``'  
(2,76): error CS1002: ; expected  
(2,76): error CS1056: Unexpected character ``'  
(2,77): error CS7017: Member definition, statement, or end-of-file expected  
(2,85): error CS1002: ; expected  
(2,85): error CS1056: Unexpected character ``'  
(2,100): error CS1525: Invalid expression term ''  
(2,100): error CS1002: ; expected  
(2,100): error CS1056: Unexpected character ``'  
(3,5): error CS1002: ; expected  
(3,20): error CS1002: ; expected  
(3,31): error CS1002: ; expected  
(3,49): error CS1002: ; expected  
(3,60): error CS1002: ; expected  
(3,67): error CS1002: ; expected  
(3,67): error CS1056: Unexpected character ``'  
(3,68): error CS7017: Member definition, statement, or end-of-file expected  
(3,70): error CS1056: Unexpected character ``'  
(3,73): error CS1056: Unexpected character ``'  
(3,77): error CS1031: Type expected  
(3,77): error CS1056: Unexpected character ``'  
(3,85): error CS1002: ; expected  
(3,85): error CS1056: Unexpected character ``'  
(3,86): error CS7017: Member definition, statement, or end-of-file expected  
(3,89): error CS1056: Unexpected character ``'

```
(4,8): error CS1002: ; expected
(4,18): error CS1002: ; expected
(4,31): error CS1002: ; expected
(4,42): error CS1002: ; expected
(4,62): error CS1002: ; expected
(4,66): error CS1002: ; expected
(4,66): error CS1056: Unexpected character '''
(4,73): error CS1519: Invalid token ''' in a member declaration
(4,73): error CS1056: Unexpected character '''
(6,13): error CS1002: ; expected
(6,14): error CS1040: Preprocessor directives must appear as the first non-
whitespace character on a line
(8,9): error CS1003: Syntax error, ',' expected
(8,10): error CS1002: ; expected
(9,3): error CS1525: Invalid expression term ''
(9,3): error CS1002: ; expected
(9,3): error CS1056: Unexpected character '''
(9,16): error CS1525: Invalid expression term ''
(9,16): error CS1002: ; expected
(9,16): error CS1056: Unexpected character '''
(9,33): error CS1002: ; expected
(9,45): error CS1002: ; expected
(9,54): error CS1002: ; expected
(10,3): error CS1525: Invalid expression term ''
(10,3): error CS1002: ; expected
(10,3): error CS1056: Unexpected character '''
(10,15): error CS1002: ; expected
(10,15): error CS1056: Unexpected character '''
(10,27): error CS1002: ; expected
```

csharp

```
public IEnumerable<int> GetNumbers()
{
    yield return 0;
    yield return 1;
    yield return 2;
    yield return 3;
    yield return 4;
    yield return 5;
}

Console.WriteLine("foreach:");
```

```
foreach (var number in GetNumbers())
{
    Console.WriteLine(number);
}

Console.WriteLine("while:");
IEnumerator<int> enumerator = GetNumbers().GetEnumerator();
while (enumerator.MoveNext())
{
    Console.WriteLine(enumerator.Current);
}
```

foreach:

0

1

2

3

4

5

while:

0

1

2

3

4

5

csharp

```
public IEnumerable<int> GetReverseNumbers(int count)
{
    for (int i = 0; i < count; i++)
        yield return i;
}

Console.WriteLine("foreach:");
foreach (var number in GetReverseNumbers(6))
{
    Console.WriteLine(number);
}
```

```
}

Console.WriteLine("while:");
IEnumerator<int> enumerator = GetReverseNumbers(6).GetEnumerator();
while (enumerator.MoveNext())
{
    Console.WriteLine(enumerator.Current);
}
```

foreach:

0

1

2

3

4

5

while:

0

1

2

3

4

5

csharp

Iný príklad:

```
(1,12): error CS1003: Syntax error, ',' expected  
(1,13): error CS1002: ; expected
```

csharp

```
public IEnumerable<int> Power(int number, int exponent)  
{  
    int result = 1;
```

```
    for (int i = 0; i < exponent; i++)
    {
        result = result * number;
        yield return result;
    }
}

// Zobrazí postupne mocniny 2^n pre ∀n∈⟨1,8⟩
// Vypíše: 2 4 8 16 32 64 128 256
foreach (int i in Power(2, 8))
{
    Console.Write("{0} ", i);
}
```

2

4

8

16

32

64

128

256

## Komplexnejší príklad:

(1,21): error CS1003: Syntax error, ',' expected

(1,22): error CS1002: ; expected

```
class MyData : IEnumerable<int>
{
    private List<int> _numbers = new List<int>();

    public void Add(int number) => _numbers.Add(number);

    // Rozhranie IEnumerable<int> implementované implicitne:
    public IEnumerator<int> GetEnumerator()
    {
        return _numbers.GetEnumerator();
        // Alebo môžeme vrátiť aj enumerátor z metódy vracajúcej IEnumerable<i
        //return GetReverse().GetEnumerator();
    }

    // Rozhranie IEnumerable implementované explicitne:
    IEnumerator IEnumerable.GetEnumerator()
    {
        return GetEnumerator();
    }

    public IEnumerable<int> GetReverse()
    {
        for (int i = _numbers.Count - 1; i ≥ 0; i--)
        {
            yield return _numbers[i];
        }
    }

    public IEnumerable<int> GetDoubled()
    {
        foreach (var number in _numbers)
```

```

        {
            yield return number * 2;
        }
    }

    public IEnumerable<int> GetEverySecondNumbers()
    {
        bool use = false;
        foreach (var number in _numbers)
        {
            use = !use;
            if (use)
                continue;

            yield return number;
        }
    }

    public IEnumerable<int> GetOnlyPositiveNumbers()
    {
        foreach (var number in _numbers)
        {
            if (number > 0)
                yield return number;
        }
    }

    public IEnumerable<int> GetFirstNumbers(int maxNumberCount)
    {
        int count = 0;
        foreach (var number in _numbers)
        {
            yield return number;

            count++;
            if (count == maxNumberCount)
                yield break; // Ukončí enumerovanie
        }
    }
}

```

```
var data = new MyData();
data.Add(1);
data.Add(-2);
data.Add(3);
data.Add(4);
data.Add(-5);

string output = string.Empty;
foreach (var d in data)
{
    output += d + " ";
}
Console.WriteLine($"Foreach: {output}");

output = string.Empty;
foreach (var d in data.OnlyPositiveNumbers())
{
    output += d + " ";
}
Console.WriteLine($"Len kladné čísla: {output}");
```

Foreach: 1 -2 3 4 -5

Len kladné čísla: 1 3 4

```
public void Display<T>(string text, IEnumerable<T> enumerable)
{
    string output = string.Empty;
    foreach (var d in enumerable)
    {
        output += d + " ";
    }
}
```

```
        Console.WriteLine($"{text}: {output}");
    }

    Display("Foreach", data);
    //Display("Foreach", data.GetEnumerator()); // Error CS150
    Display("Opačné poradie", data.GetReverse());
    Display("Vynásobené číslom 2", data.GetDoubled());
    Display("Každé druhé číslo", data.GetEverySecondNumbers());
    Display("Len kladné čísla", data.GetOnlyPositiveNumbers());
    Display("Prvé tri čísla", data.GetFirstNumbers(3));
```

Foreach: 1 -2 3 4 -5

Opačné poradie: -5 4 3 -2 1

Vynásobené číslom 2: 2 -4 6 8 -10

Každé druhé číslo: -2 4

Len kladné čísla: 1 3 4

Prvé tri čísla: 1 -2 3

csharp

## Odkazy

Microsoft dokumentácia:

- [Rozhrania](https://learn.microsoft.com/en-us/dotnet/csharp/fundamentals/type)
- [Klíčové slovo interface](https://learn.microsoft.com/en-us/dotnet/csharp/la

(1,1): error CS1024: Preprocessor directive expected  
(2,23): error CS1003: Syntax error, ',' expected  
(2,24): error CS1002: ; expected  
(4,12): error CS1003: Syntax error, ',' expected  
(4,18): error CS1003: Syntax error, ']' expected  
(4,18): error CS1026: ) expected  
(4,18): error CS1002: ; expected  
(4,27): error CS1001: Identifier expected  
(4,27): error CS1514: { expected  
(4,27): error CS1513: } expected  
(4,27): error CS7017: Member definition, statement, or end-of-file expected  
(4,115): error CS1733: Expected expression  
(4,115): error CS8124: Tuple must contain at least two elements.  
(4,115): error CS1026: ) expected